P-L-130C
September 28, 1976
SUPERSEDING
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January 14, 1975

FEDERAL SPECIFICATION

LAUNDRY SOUR
(BLUING AND BRIGHTENER)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers laundry sours with bluing agent or optical brightener intended for use in laundries (see 6.1).

1.2 Classification. Laundry sours shall be of the following types and classes, as specified (see 6.2).

Type I - With bluing agent.

Class 1 - Deep tint.
Class 2 - Medium tint.
Class 3 - Light tint.

Type II - With optical brightener.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

PPP-P-45 - Packaging of Bulk Quantities of Soaps, Detergents and Related Products In Chip, Powdered or Granular Form.

Federal Standard:

Fed. Test Method Std. No. 191 - Textile Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

16 CFR 1500 - Federal Hazardous Substances Act.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

E 70 - pH of Aqueous Solutions with the Glass Electrode.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Association of Official Analytical Chemists (AOAC):

Official Methods of Analysis of the Association of Official Analytical Chemists - Eleventh Edition, 1970.

(Application for copies should be addressed to the Association of Official Analytical Chemists, P.O. Box 540, Benjamin Franklin Station, Washington, DC 20044.)

American Association of Textile Chemists and Colorists (AATCC):

The Color Index, Vols. 1-5

(Application for copies should be addressed to the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 Type I. Type I laundry sour shall be a clean, uniform, free flowing, granulated powder containing a mixture of the high neutralizing laundry sour component specified in 3.1.4 and the laundry blue component specified in 3.1.5. The two components shall be mixed in the proportions specified in table I. A solution of the laundry blue component in water shall be sprayed onto the laundry sour component in a suitable mixer.

TABLE I. Ounces of bluing agent per 100 pounds of product

Class	
1	4
2	2
3	1

3.1.2 Type II. Type II laundry sour with optical brightener shall be a clean, uniform, free flowing, granulated powder containing a mixture of the high neutralizing laundry sour component specified in 3.1.4 and a fluorescent optical brightener specified in 3.1.6.

3.1.3 Excluded ingredients. Type I and type II laundry sours shall contain no boric acid or boric acid derivatives.

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3.1.4 Laundry sour component. The laundry sour component shall consist of not less than 60.0 percent sodium fluosilicate.

3.1.5 Laundry blue component. The laundry blue component shall be a mixture of 100 percent water-soluble dyes of the colors and in the proportions specified in table II.

TABLE II. Laundry blue component composition

Color index #	Weight percent	Examples of suitable dyes
Color index 42650	56	Acid Violet 6B Conc.-Ciba-Geigy Corp. Acid Violet 4BN Conc.-Sandoz Company
Color index 63000	44	Erio Fast Cyamine S Conc.-Ciba-Geigy Corp. Alizarine Light Blue SE-Sandoz Company

3.1.6 Fluorescent optical brightener. The optical brightener shall be water soluble, effective over a wide pH range, and suitable for use in all laundries. Type II laundry sour shall produce a brightening effect equal to or greater than the reference sour specified in 4.3.8.2 when tested as specified in 4.3.8.

3.2 Tint. Type I laundry sour shall tint clothes uniformly when tested as specified in 4.3.3 and the tint shall be removed by washing when tested as specified in 4.3.4.

3.3 Alkali neutralization capacity. It shall take not less than 500 mg of NaOH per gram of laundry sour to adjust the pH to 7.0, when tested as specified in 4.3.5.

3.4 Solubility. Ninety-five percent of the types I and II laundry sours shall dissolve in water at 45°C (113°F) within 2 minutes and all of the laundry blue component in type I laundry sour shall dissolve within 2 minutes, when tested as specified in 4.3.6.

3.5 pH value. The pH value of a 0.2 percent (by weight) distilled water solution of types I and II laundry sours shall be not more than 3.60, when tested as specified in 4.3.7.

3.6 Labeling (all types). The containers shall be labeled in accordance with the Federal Hazardous Substances Act. Each container of laundry sour of specified type shall be durably and legibly marked with the following directions for use and precautions.

3.6.1 Use label. All products shall be labeled with the following directions for use information:

DIRECTIONS FOR USE: For general souring add sufficient laundry sour to the washwheel of the normal souring operation to attain the pH values listed below. In most instances, only 1 to 2 ounces of product per 100 pounds of load are necessary to achieve the recommended pH values. High bicarbonate water and high concentrations of residual alkalinity in the rinse bath will necessitate additional amounts to achieve the desired pH. Determine the exact amount required by using the washroom control kit (sour indicator).

Classification	Desired color with Sour Indicator	pH
Hospital and commercial flat	yellow	6.0 - 6.5
Family flat and wearing apparel	yellowish-orange	5.2 - 6.0
Starch work and diapers	orange-orange red	4.5 - 5.2

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3.6.2 Warning label. All products shall be labeled in accordance with the Federal Hazardous Substances Act and with the following warning information.

WARNING: INJURIOUS TO SKIN AND EYES
HARMFUL IF SWALLOWED

Contains fluoride salts. Avoid contact with eyes and mucous membranes and prolonged contact with skin.
In case of external contact, flush with water.
For eye contact, flush with water for 15 minutes.
Get prompt medical attention.
If swallowed, give large quantities of water or milk.
Call a physician.
Keep out of reach of children.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Examination of the end item. The end item shall be examined in accordance with the classification of defects, inspection levels, and acceptable quality levels (AQL's) set forth below. The lot size, for purposes of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of containers, as specified, of the applicable size, type, and class material contained herein.

4.2.2 Examination of the end item container for defects in appearance of laundry sour. Defects for the end item shall be examined and scored as listed in table III. The sample unit for this examination shall be one filled shipping container fully prepared for delivery. Sampling and inspection shall be in accordance with MIL-STD-105, inspection level S-2 with an AQL of 4.0 expressed in terms of percent defective.

TABLE III. Examination of the end item

Examine	Defect
Contents	Not uniform color, not thoroughly mixed, not free-flowing.
Net weight	Less than specified.

4.2.3 Examination of preparation for delivery. The packing and marking shall be examined in accordance with PPP-P-45.

4.2.4 Testing of the end item. The end item shall be tested for conformance to the requirements of section 3, in accordance with the test methods specified in table IV. The sample unit, for purposes of testing, shall be a one pound composite sample selected in accordance with 4.2.4.1. The average of duplicate determinations shall meet the requirements. Both determinations shall meet the requirements when the test is applicable to the individual unit. The lot size shall be expressed in units of containers of the specified quantity, type, and class of laundry sour. The sample unit shall be tested with no evidence of failure to meet the applicable requirements. Test reports shall contain the individual values utilized in expressing the final results. The lot shall be unacceptable if the composite fails to meet any test requirements specified.

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4.2.4.1 Composite sample. A sample of not less than 1 pound each, shall be taken from not less than four separate shipping containers, selected throughout the lot presented for test. The total sample shall consist of not less than three sample portions taken at random from each separate shipping container, and shall be placed in a container of a suitable size. After thorough mixing, a 1 pound sample shall be drawn for testing purposes. The composite sample shall be placed in an airtight container, sealed and marked with the identifying lot number and date.

TABLE IV. Instructions for testing end item

Characteristics	Specification reference: Requirement	Specification reference: Test method	No. determination per composite sample unit	Results reported as: Pass or fail 1/	Results reported as: Numerically to nearest 2/
Sodium fluosilicate	3.1.4	4.3.1	2		1 percent
Color, type I	3.1.5	4.3.2	2	X	
Tint uniformity, type I	3.2	4.3.3	2	X	
Tint washability, type I	3.2	4.3.4	2	X	
Alkali neutralization capacity	3.3	4.3.5	2		1 mg
Solubility	3.4	4.3.6	Average of 2	X	
pH value	3.5	4.3.7	Average of 2		0.1 pH
Fluorescent optical brightener	3.1.6	4.3.8	2	X	

1/ Test report shall include all values on which results are based.
2/ If failure is indicated, report either description of failure or numerical point of failure, as applicable.

4.3 Test methods.

4.3.1 Sodium fluosilicate (all types). Determine sodium fluosilicate in laundry sour in accordance with the AOAC Method for "Fluorine Present as Sodium Fluosilicate" (Eleventh Edition, 6.024 and 6.025).

4.3.2 Color, type I.

4.3.2.1 Color standard. Prepare the color standard by mixing the following dyes in the following proportions:

56% - Color Index No. 42650 (Acid Violet 6B Conc.-Ciba-Geigy Corp.)
44% - Color Index No. 63000 (Erio Fast Cyanine S Conc.-Ciba-Geigy Corp.)

Mix the two dyes together thoroughly and prepare a solution containing 2.5 mg of standard color dye mixture per liter of distilled water.

4.3.2.2 Color comparison. Dilute the filtrate from 4.3.6 to 1 liter. Using Nessler tubes, visually compare equal volumes of the filtrate with: The color standard from 4.3.2.1 for class 1 sour and blue; a 1:1 dilution of the color standard in water for class 2 sour and blue, or 1:3 dilution of the color standard in water for class 3 sour and blue. The colors shall have the same hue and intensity.

4.3.3 Tint uniformity, type I. Place a 7 inch by 7 inch piece of white sheeting in a suitable stainless steel cylindrical container (see Fed. Test Method Std. No. 191, Test Method No. 5610, Colorfastness to Laundering of Cotton and/or Linen Textile Materials; Launder-Ometer Method) and add 50 ml of a 0.1 percent solution by weight of laundry sour in distilled water to the container. Agitate the container and contents for 15 minutes at room temperature in a Launder-Ometer as specified in Test Method 5610. Remove the cloth from the container, extract the excess fluid, and press the cloth as specified in Test Method 5610. The tinting shall be uniform, not streaked or spotty.

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4.3.4 Tint washability, type I. Wash and press the cloth following the laundering and pressing instructions in Fed. Test Method Std. No. 191, Test Method No. 5550, Shrinkage in Laundering; Cotton, Linen and Mixed Cotton and Linen Cloth. No traces of tint shall remain.

4.3.5 Alkali neutralization capacity (all types). Take a sample of approximately 1.0 g of laundry sour, record the weight to the nearest 0.01 g, and dissolve in 500 ml of CO_2 -free distilled water in a 1000 ml beaker. Place glass and calomel electrodes of a calibrated pH meter and mechanical stirrer in beaker. Start stirrer. Titrate the solution to a pH of 7.0 with 0.5N NaOH. Use care near the end-point due to a lag in the pH meter. Calculate the alkali neutralization capacity of the laundry sour and blue as follows:

$$\text{Alkali neutralization capacity} = \frac{A \times B \times 40}{W_1}$$

where: A = ml of NaOH solution used.
B = normality of NaOH solution.
W_1 = weight of sample in grams.

4.3.6 Solubility (all types). Accurately weigh a 1 gram sample of thoroughly mixed laundry sour. Place in a 1 liter beaker and add 500 ml of distilled water at 45°C (113°F). Stir moderately for 2 minutes and filter through a piece of tared, white, quantitative filter paper in a Buchner funnel using moderate suction. Wash the precipitate with 250 ml of distilled water at 45°C (113°F). Save the filtrate for color determination (see 4.3.2). Dry the filter paper to constant weight in a vacuum at 60°C and weigh it to determine the weight of laundry sour that did not dissolve. Report the percent of the sample that did not dissolve. There shall be no dark colored specks of undissolved blue component in the type I laundry sour, on the filter paper.

4.3.7 pH value (all types). Make a 0.2 percent solution of the laundry sour in freshly boiled and cooled (room temperature) distilled water, as specified in 3.5, and mix. Determine the pH value of this solution in accordance with ASTM E 70.

4.3.8 Optical brightening agents.

4.3.8.1 Equipment and reagents.

a. Terg-O-Tometer or equal.
b. Bleached and unbrightened fabric cotton test swatches - Test Fabrics, Inc., Cat. No. 400, Middlesex, NJ.
c. Macbeth Lamp Model AKP-82 or equal - Macbeth Corporation, Newburgh, New York.
d. Tinopal CBS Flourescent Brightener or equal - Ciba-Geigy Corporation.
e. Sodium fluosilicate - Olin Corporation or Ashland Chemical.
f. Sodium chloride or sodium sulfate, reagent grade.

4.3.8.2 Sour formulation (weight percent).

Sodium fluosilicate	60.00
Tinopal CBS	0.80
Sodium chloride/sodium sulfate	39.20

4.3.8.3 Procedure. The laundry sour shall be evaluated against the laboratory prepared reference sour formulation specified in 4.3.8.2. Wash tests shall be performed in a Terg-O-Tometer (or equal) in a liter of 1:4000 wt/wt laundry sour at 100° ± 2°F. Cotton swatch (4 x 5 inch) shall be introduced into the solution and washed for 5 minutes at a speed of 120 cycles per minute. The swatch shall be rung out and pressed dry between two pieces of clean, white sheeting. The reference swatch shall be treated in the same way with the reference sour formulation. The swatches shall then be evaluated visually under the lights of the Macbeth lamp or the light source from the north sky daylight to determine if the effect obtained with the laundry sour is equal to that obtained with the applicable reference laundry sour.

5. PREPARATION FOR DELIVERY

5.1 Packing. The laundry sour shall be packed in accordance with PPP-P-45, in the level, container and quantity specified (see 6.2).

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5.2 Marking. The containers shall be marked in accordance with 3.6 and PPP-P-45.

6. NOTES

6.1 Intended use. The laundry sours covered by this specification is intended for use in laundries to neutralize any residual alkali that may be present in fabrics after completion of rinsing cycle, and to improve the whiteness of the laundered texiles.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and class required (see 1.2).
(c) Level of packing required (see 5.1).
(d) Quantity and container required (see 5.1).

MILITARY INTERESTS:

Custodians:

Army - GL (MCA)
Navy - SH
Air Force - 84

Review activity:

Army - MD

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS
NBS
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CPSC

PREPARING ACTIVITY: GSA - FSS

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